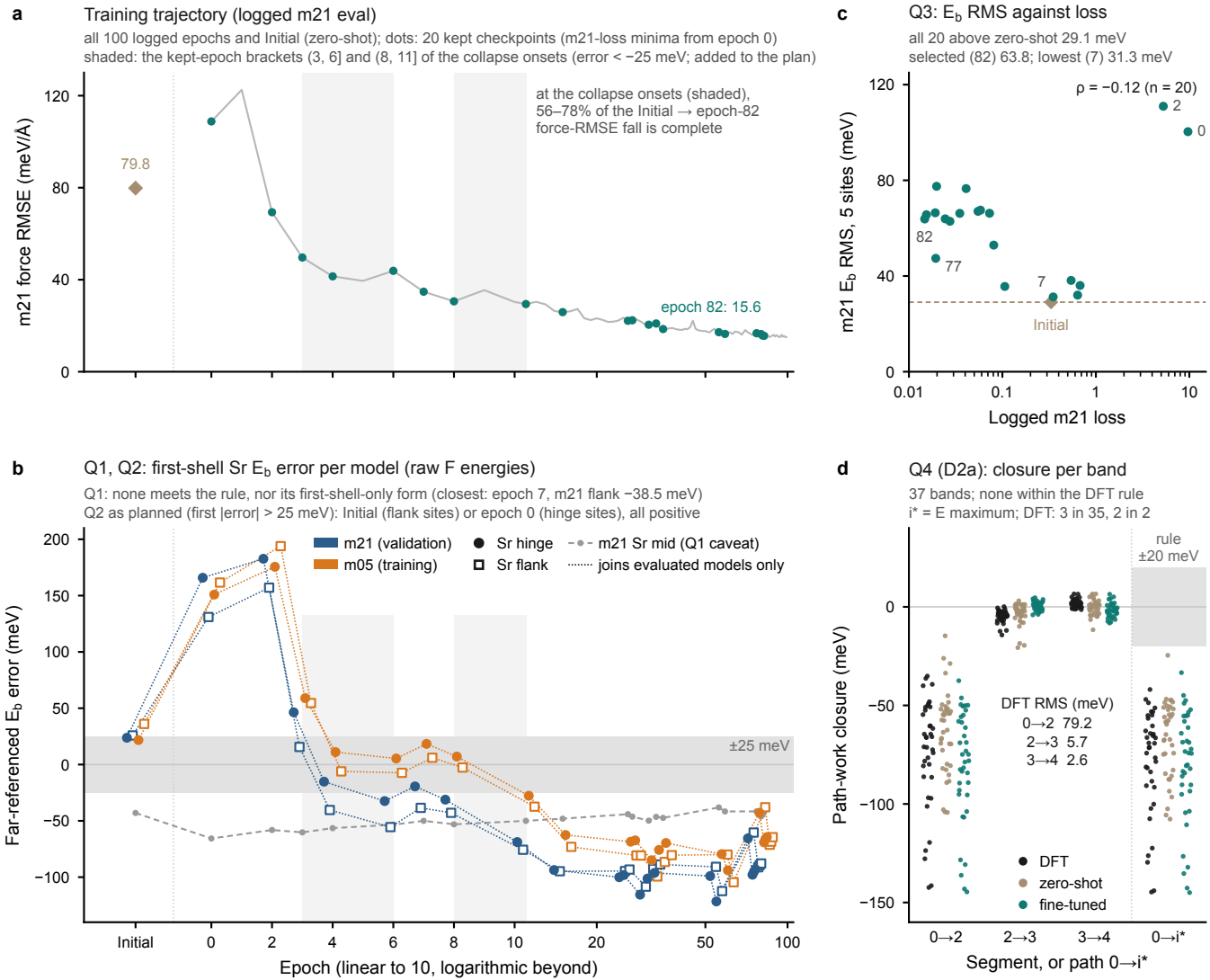

a
Training trajectory (logged m21 eval)
all 100 logged epochs and Initial (zero-shot); dots: 20 kept checkpoints (m21-loss minima from epoch 0)
shaded: the kept-epoch brackets (3, 6] and (8, 11] of the collapse onsets (error < −25 meV; added to the plan)
m21 force RMSE (meV/Å)
120
80
40
0
79.8
at the collapse onsets (shaded),
56–78% of the Initial → epoch-82
force-RMSE fall is complete
epoch 82: 15.6
c
Q3: E_b RMS against loss
all 20 above zero-shot 29.1 meV
selected (82) 63.8; lowest (7) 31.3 meV
ρ = −0.12 (n = 20)
2
0
82
77
7
Initial
m21 E_b RMS, 5 sites (meV)
120
80
40
0
0.01
0.1
1
10
Logged m21 loss
b
Q1, Q2: first-shell Sr E_b error per model (raw F energies)
Q1: none meets the rule, nor its first-shell-only form (closest: epoch 7, m21 flank −38.5 meV)
Q2 as planned (first |error| > 25 meV): Initial (flank sites) or epoch 0 (hinge sites), all positive
m21 (validation)
m05 (training)
Sr hinge
Sr flank
m21 Sr mid (Q1 caveat)
joins evaluated models only
Far-referenced E_b error (meV)
200
150
100
50
0
−50
−100
±25 meV
Initial
0
2
4
6
8
10
20
50
100
Epoch (linear to 10, logarithmic beyond)
d
Q4 (D2a): closure per band
37 bands; none within the DFT rule
i* = E maximum; DFT: 3 in 35, 2 in 2
rule
±20 meV
Path-work closure (meV)
0
−50
−100
−150
DFT RMS (meV)
0→2 79.2
2→3 5.7
3→4 2.6
DFT
zero-shot
fine-tuned
0→2
2→3
3→4
0→i*
Segment, or path 0→i*